Theory Assignment 1:

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* Examine on linearity, time variance, causality and stability of the following signal with appropriate mathematical analysis.

$$y(n) = x(n)\, u(n-10)$$

Linearity :

Let,

$$y_1(n) = x(n)\, u_1(n-10) \quad \text{— (I)}$$

$$y_2(n) = x(n)\, u_2(n-10) \quad \text{— (II)}$$

be the systems outputs to arbitrary inputs $u_1(n-10)$ and $u_2(n-10)$. Then with, $u_3(n-10)$

$$\cancel{\alpha_1 u_1[n-1]} = \alpha_1 u_1(n-10) + \alpha_2 u_2(n-10)$$

we get,

$$y_3(n) = x(n)\, u_3(n-10)$$

$$= x(n)\left(\alpha_1 u_1(n-10) + \alpha_2 u_2(n-10)\right)$$

$$= x(n)\, \alpha_1 u_1(n-10) + x(n)\, \alpha_2 u_2(n-10).$$

As it satisfies the superposition theory, the system can be called linear.

Time - variance :

Let $u_2(n-10) = u_1(n+n_0-10)$

where $n_0 \neq 0$ for all n, then

$$y_1(n+n_0) = x(n+n_0)\, u_1(n+n_0-10)$$

$$= x(n+n_0)\, u_2(n-10)$$

But, $x(n+n_0)\, u_2(n-10) \neq x(n)\, u_2(n-10)$

Because a time shift of n_0 in the input does not correspond to a time shift of n_0 in the output, the system ~~is not~~ can be called time variant.

Causality :

$$y(n) = x(n)\, u(n-10)$$

In this equation we can see that the output doesnot depend on future input values, so it can be called causal.

Stability :

Let us assume $u(n-10)$ is a bounded input. ~~Then~~ But $y(n) = x(n)\, u(n-10)$, where the output grows without bound despite a bounded input. The system is therefore not stable.